

BAND: 1

PROGRAM:

BACHELOR OF SCIENCE DEGREE IN SOFTWARE ENGINEERING

Compiled By:

1. Girma Neshir, Ph.D, Assistant Prof., Addis Ababa Science and Technology University(AASTU)
2. Birhanu Sisay, M.Sc. Lecturer, Arba Minch University(AMU)

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Table of Contents

1. Introduction 2
2. Objective of test blueprint 3
3. Steps to develop test blueprint 3
4. Issues to consider in test blueprint preparation 4
5. Required documents to prepare the test blueprint 4
   i. **Knowledge** 4
   ii. **Skill** 5
   iii. **Attitude** 6
6. Conclusion 20

1. Introduction

The Ministry of Education of Ethiopia has announced the implementation of exit exam for all undergraduate program students (public and private), beginning with the 2023/2024 academic year, in order to improve the quality of graduates produced by higher learning institutions. The exit exam aimed at checking whether students have acquired the required knowledge, skills and attitudes or not. To implement this, it requires determining competency areas for a specific program, which is already completed. Based on the competency areas prepared, it needs to plan the construction of tests.

Planning of a test is a first and vital step in the construction of an achievement test. An achievement test demands very systematic and careful planning, as a fact that good planning is a symbol of success. Tests are the tools that provide scores that measure level of student learning and study program learning outcomes. In order to achieve the valid and reliable measurement of student learning and program learning outcomes, the development of valid and reliable test is the mandatory. Test should be able to measure student performance in all dimensions of knowledge, skill and attitude. The carefully planned test construction contributes to improve the overall quality of the test in terms of test content validity, difficulty level, discrimination power and test reliability. Test preparation is not an easy task; it requires a careful planning and guideline to make the task simple. Test construction needs the preparation of test blue print. Test blueprint is defined as a complete plan that explains how to develop a test. The term refers to a map or specification of assessment to ensure that all aspects of the curriculum and educational domains are covered by the assessment programs over a specified period of time. It helps curriculum developers/test constructors to match various competencies with the course content and the appropriate modality of assessment.

Generally, test blueprint will help to ensure tests: 1) Appropriately assess the achievement of instructional objectives of the course; 2) Appropriately reflect key course goals, objectives and the material learned or covered during the instruction period; and 3) Include the appropriate item formats along with the knowledge and skills being assessed.

Keeping this in mind, the team has prepared this test blueprint document in order to help the test developers or content specialists in their process of valid and reliable test construction. The major points considered in the process of preparing this test blue print guideline were the core competencies that have been already identified for the themes of courses, the course contents, course credit hours, and the learning outcomes with their corresponding levels of achievement by learning domains. In line with these, the number of test items that should adequately assess the performance of students in all the content topics will be determined through discussion with the content specialists who construct the blueprint and test for their corresponding study program.

Therefore, the main purpose of this manual is to give direction on how to develop blueprint for content specialists so that they can develop a test blueprint for their respective program.

2. Objective of test blueprint

Test blueprint preparation is generally opted to assist the preparation of a test that is representative, broadly sampled, and consisting of complete knowledge domain expected of the Ethiopian higher education students on completion of their study program. The specific objectives of test blueprint are to:

- Facilitate the construction of a representative and balanced test items for the selected courses in accordance with the competencies identified.
- Guide test developers or writers to write or set appropriate test items.

3. Steps to develop test blueprint

i. Listing the identified core competencies corresponding to the respective course
ii. Listing the specific and minimum competencies to be assessed by the exam
iii. Listing out the major learned course contents corresponding to these listed minimum competencies.
iv. Listing the specific, measurable learning outcomes under these minimum competencies using action verbs (Note: the number and the importance of these learning outcomes proportionally contribute to the total test items. The weight of the categories of the list of learning outcomes under each competencies and contents should lead to the number of total test items to be constructed)

v. Determination of the test format
vi. Content weighting - Experts judge the weight to be given to the listed learning outcomes and their corresponding contents.
vii. Identifying the categories of learning domains to which each of these learning outcomes belongs using their action verbs.
viii. Construction of test blueprint table
ix. Allocating the number of test items for each learning outcomes by their corresponding types and levels of learning domains.
x. Determination of the sub-total and total test items by their categories of learning domains and learning outcomes.

4. Issues to considered in test blueprint preparation

i. Curriculum details
ii. Course outlines/guidebook details, lists of learning outcomes, lecture notes, textbooks, and other curricular materials
iii. Course contents pertinent to the respective learning outcomes
iv. Time devoted to cover the content with in a course (course credit hour)
v. Total test length (i.e., total number of test items)
vi. Number of questions allocated to each content area using the proportion calculated
vii. Weight for each item
viii. Relevance of the content to students future career

In addition to these, following the template mentioned below will help to prepare the blueprint.

5. Required documents used for the test blueprint

To prepare a test blueprint, we used the following documents in our hand.

1. Identified core competencies and learning outcome of the program

Software engineering graduates are expected to demonstrate the following knowledge, skill and attitudes:

i. Knowledge

- Conduct investigation of complex problems using research-based knowledge and research methods including design of experiments, analysis and

interpretation of data, and synthesis of information to provide valid conclusions

- Adopt, innovate, select, and apply appropriate state of the art software design and development tools, methodologies, resources, and emerging technologies including simulation, prediction and modeling for complex business process with understanding of their potential capabilities, limitations and associated risks.
- Identify, formulate, research literature, and analyze complex problems, reaching substantiated (justifiable, validated) conclusions using software development principles, methodologies, and tools.
- Apply logical reasoning and informed decision to assess societal, health, safety, privacy, security, legal and cultural issues, and the consequent responsibilities relevant to software engineering application
- Understand the impact of software applications in societal and environmental contexts and apply the principle of green computing for sustainable development.

ii. Skill

- Design integrated and efficient software solutions, component, or processes to address complex problems and implement them to meet the specified needs with appropriate consideration for public health and safety, cultural, societal (local and international) and environmental considerations.
- Communicate effectively on complex business and system activities with the software engineering community and with a society at large; such as being able to comprehend and write effective reports, articles and design documentations using various tools, make effective presentations and give and receive clear instructions.
- Demonstrate knowledge and skill on software project management principles and apply these to manage software development processes and its resource requirements, costs in multidisciplinary projects
- Apply knowledge of mathematics, and software engineering specialization to address complex and large-scale problems.

iii. Attitude

- Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.
- Function effectively as an independent software engineer, as a member or as a leader in a diverse team and multidisciplinary settings.
- Recognize the need for and have the preparation as well as ability to engage in an independent and lifelong learning in the information and knowledge economy.

2. Curriculum of the program

   See Appendix B of Curricula of BSc in Software Engineering of Addis Ababa Science and Tech. University and Arba Minch University

- **Expected profile of the Graduates**

This exit exam is expected to measure the software engineering graduates knowledge, skill and attitude so that they:

- Use current techniques, skills, and tools necessary for software development, testing, and maintenance.
- Acquire the knowledge and innovative skills to plan, set up and run software related businesses.
- Understand best practices and standards of software engineering program and their application.
- Be familiar with the required software project management skill that enable them to manage complex software projects
- Understand professional, ethical, legal, security and social issues and responsibilities.
- Have ability to apply knowledge of computing and mathematics to resolve on-hand and upcoming problems.
- Have recognition of the need for, and an ability to engage in, continuing professional development and the knowledge and skills to act as research assistants or lecturers in higher education institutions

- **General objective**

  The software engineering program shall have the following general objectives:

- To produce large scale software developers, project managers and leaders who has an ability to address complex and large-scale problems.
- To produce digital innovators and entrepreneurs.
- To nurture professionally qualified software engineers with the potential to become consultant, trainer, software requirement engineer, software architect, system analyst, software quality assurance and tester.
- To produce software engineers who incorporate standards and security in software development with appropriate consideration of socio-cultural and environmental safety.
- To produce researchers and software engineers who will develop dynamic and smart software solutions using emerging technologies to address complex national and global challenges and demonstrate lifelong learning and continuing professional development.

- **Specific objectives/selected from courses specific objectives**

By the end of the program, the graduates will be able achieve the following specific objectives:

- Identify the basic concepts of programming with its program development process
- Apply the basic programming constructors' and the various programming techniques
- Demonstrate problem solving skills by analyzing a variety of real-world problems using program solving techniques
- Analyze modular programming, user defined data type; file stream handling and vectors by solving variety programming problems.
- Develop, debug and test medium scale applications by analysing a variety of real-world problems
- Follow the ability of self-education and life-long learning needed by computer programming
- State how data is organized in a computer, how it can be retrieved, and how it can be used
- Explain the basic techniques for the design and analysis of efficient Algorithm
- Apply various data structures including Linked lists, Queues, Stacks, Binary trees, Search trees and Graphs

- Distinguish complexity, efficiency of searching and sorting algorithms using Empirical and Theoretical analysis
- Design algorithms to solve real-life problems
- Discuss major concepts of object-oriented programming, object technology and programming environment.
- Use inheritance, encapsulation, polymorphism exception handling and I/O functionality
- Create event-driven GUI containers and components within an integrated development environment
- Design and develop object-oriented software
- Recognize a web protocol, design and development of web-based application
- Use modern tools and techniques for developing web-based applications
- Develop a websites, web frameworks using basic HTML, CSS, JavaScript, AJAX and Server-Side scripting
- Create dynamic and interactive web pages as well as ability to connect to a wide range of databases
- Assess web-based solution and engage in an independent and lifelong learning in the broad scope of Internet programming as well as web application
- State the basic components of an Android application and mobile computing
- Demonstrate basic skills mobile computing of using an integrated development environment
- Evaluate the security concerns of the mobile computing technologies and mobile networks.
- Choose, the legal and ethical principles mobile application framework for designing, and constructing mobile Software application
- Explain fundamentals of database system concepts, technology, models of database and optimization.
- Write SQL statements for data creation and manipulation purposes
- Design the database models from specifications using a popular DBMS
- Manage the possible problems that may occur in securing data.
- Recall the structure and functionality of modern Operating Systems
- Explain conditions that lead to deadlock and memory management, I/O, mass storage and file management issues
- Describe security and protection measures used in operating systems
- Use knowledge of process management and synchronization for software development

- Compare the various algorithms on which the core functions of the Operating Systems are built on
- Identifies the basic principles of Software Engineering
- Describe software process and state-of-the-arts software process models
- Explain fundamental concept of data communication, computer networks, and computer system security as well as setup and configure small-area network.
- Demonstrate installation and configuration of system software, and make use of software engineering tools.
- Be aware of the professional, ethical, and norm of software engineering practice.
- Describe the role of requirement engineering in software development
- Use the appropriate requirements elicitation techniques to identify requirements.
- Develop effective functional and non-functional requirements document that are complete, concise, correct, consistent, testable and unambiguous.
- Manage the change to software requirements.
- Explain the role of software architecture and software design principles
- Apply the different types of design patterns in software development
- Evaluate various architectural styles and solutions based on quality attributes
- Design a large-scale software system that reflects and balances the different needs of its stakeholders
- Explain software project management tasks
- Develop work breakdown structure, schedule and cost for software projects.
- Use project management tools, techniques, and skills
- Prepare plan documents for software project management, risk management, software quality management and assurance
- Apply modern software testing processes in relation to software development and project management
- Create test strategies and plans, design test cases, prioritize and execute them.
- Manage incidents and risks within a project
- Become competent expert in designing, implementation and development of Quality software systems and IT processes
- Examine the major software defects, bugs and code smells
- Estimate the maintenance cost, efforts, size and time for software maintenance.
- Apply the best practices and models in software maintenance process
- Maintain a software based on the reengineering and refactoring methods and principles.

- Describe concepts of network fundamentals.
- Analyze the different networking layers
- Design appropriate computer network services based on a given requirement
- Identify malicious code functions, threats, vulnerabilities possible propagation, and mitigation methods
- Assess computing cryptography and cryptanalysis
- Develop mechanisms for authentication and access control in a domain.
- Compare types of firewall and intrusion detection methods
- Explain fundamental concepts of artificial intelligent agents
- Compare the basic search strategies used in an intelligent agent.
- Apply the law of logic to represent knowledge for an expert system
- Engage in independent and lifelong learning with the broad application area of Artificial Intelligence.
- Explain the distinction between supervised and unsupervised learning, as well the interests and difficulties of both approaches.
- Implement the major machine learning algorithms for data analytics
- Formulate research on a personal project using machine learning techniques

- **Themes and list of courses**

To help demarcate the focus areas and competency boundaries that are covered by the identified courses, the courses have been categorized under five themes. Accordingly, Table 1 maps courses to the respective themes identified.

Table 1: Courses categorization based on themes

S/N	Themes	Courses to be included in the Exit Exam	Cred.hrs.
1	Problem analysis and programming	1. *Fundamental of Programing	8
		2. Fundamental of Data structure and Algorithm	4
		3. Object Oriented Programming	3
		4. Internet Programming (or Web Programming)	6
		5. Mobile Application Development	3
		6. Fundamentals of Database Systems	4
		7. Operating Systems	4
2	Software requirement, Design, and Architecture	8. Fundamentals of Software Engineering	4
		9. Software Requirement Engineering	3
		10. Software Architecture and Design	3
3	Software project management, maintenance and quality assurance	11. Software Project Management	3
		12. Software Testing and Quality Assurance	3
		13. Software Evolution and Maintenance	3

4	Networking and Software Security	14. *Fundamentals of Networking, and Software and information security	7
5	AI and related Areas	15. *Fundamentals of Artificial Intelligence and Machine Learning	6
		Total Credit Hours	**64**

- **Total credit hour of the selected courses**
  - 64 credit hours
- **Share of the Themes/Courses/Items in Percentage (%) (Please see Table 2 in the appendix A)**
  - Share of themes (T) = $\frac{a}{b}\ x100$, Where "a" is the credit hour of a theme and "b" is the total credit hour of the program. Credit hour of a theme is the sum of credit hours of courses in the theme.
  - Share of courses per theme (C) = $\frac{Credit\ hour\ of\ the\ course}{Credit\ hour\ of\ the\ theme}\ x100$
  - Share of items per course = $Share\ of\ the\ course\ X\ Total\ number\ of\ items$
- **Learning outcomes in terms of the three domains (Cognitive, Affective and Psychomotor)**
- **Test blueprint Table** (Table of Specification) **(See Table 2 below)**

Ministry of Education (Table 2)
Higher Education Sub-sector
Test Blueprint (Table Specification) for BSC in Software Engineering

Themes / Share of the themes in %	General objective /competencies	Name of Courses /Share of the course in %	Credit hour	Specific objectives/Learning outcomes	No Item per Learning Outcome	Learning outcomes								No. of items
						Cognitive						Affective	Psychomotor	
						Remembering	Understanding	Application	Analysis	Evaluation	Creation/Synthesi			
Problem analysis and programming =31/64 =49%	= Conduct investigation of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions = Apply knowledge of mathematics, and software engineering specialization to address complex and large-scale problems	Fundamental of Programing =8/31*49% =11 Items	8	• Identify the basic concepts of programming with its program development process	1		1							11
				• Apply the basic programming constructors' and the various programming techniques	3			3						
				• Demonstrate problem solving skills by analyzing a variety of real-world problems using program solving techniques	2						2			
				• Analyze modular programming, user defined data type; file stream handling and vectors by solving variety programming problems.	2				2					
				• Develop, debug and test medium scale applications by analysing a variety of real-world problems	2						2			
				• Follow the ability of self-education and life-long learning needed by computer programming	1							1		
	= Conduct investigation of complex problems using research-based knowledge and research methods including	Fundamental of Data structure and Algorithm =4/31*49% =6 items	4	• State how data is organized in a computer, how it can be retrieved, and how it can be used	1	1								
				• Explain the basic techniques for the design and analysis of efficient Algorithm	1		1							
				• Apply various data structures	2			2						

	design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions = Design integrated and efficient software solutions, component, or processes to address complex problems and implement them to meet the specified needs with appropriate consideration for public health and safety, cultural, societal (local and international) and environmental considerations.			including Linked lists, Queues, Stacks, Binary trees, Search trees and Graphs										6
				• Distinguish complexity, efficiency of searching and sorting algorithms using Empirical and Theoretical analysis	1				1					
				• Design algorithms to solve real-life problems	1						1			
	= Adopt, innovate, select, and apply appropriate state of the art software design and development tools, methodologies, resources, and emerging technologies including simulation, prediction and modeling for complex business process with understanding of their potential capabilities, limitations and	Object Oriented Programming =3/31*49% = 5items	3	• Discuss major concepts of object-oriented programming, object technology and programming environment.	1		1							5
				• Use inheritance, encapsulation, polymorphism exception handling and I/O functionality	2			2						
				• Create event-driven GUI containers and components within an integrated development environment	1						1			
				• Design and develop object-oriented software	1						1			

	= Adopt, innovate, select, and apply appropriate state of the art software design and development tools, methodologies, resources, and emerging technologies including simulation, prediction and modeling for complex business process with understanding of their potential capabilities, limitations and associated risks. = Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.	Internet Programming (or Web Programming) =6/31*49% =9	6	• Recognize a web protocol, design and development of web-based application	2		2							9
				• Use modern tools and techniques for developing web-based applications	2			2						
				• Develop a websites, web frameworks using basic HTML, CSS, JavaScript, AJAX and Server-Side scripting	2						2			
				• Create dynamic and interactive web pages as well as ability to connect to a wide range of databases	2						2			
				• Assess web-based solution and engage in an independent and lifelong learning in the broad scope of Internet programming as well as web application	1					1				
	= Adopt, innovate, select, and apply appropriate state of the art software design and development tools, methodologies, resources, and emerging technologies including simulation, prediction and modeling for complex business process with understanding of	Mobile Application Development =3/31*49% = 5items		• State the basic components of an Android application and mobile computing	1	1								5
				• Demonstrate basic skills mobile computing of using an integrated development environment	2			2						
				• Evaluate the security concerns of the mobile computing technologies and mobile networks.	1					1				
				• Choose, the legal and ethical principles mobile application framework for designing, and constructing mobile Software application	1							1		

	their potential capabilities, limitations and associated risks. = Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.														
	= Apply logical reasoning and informed decision to assess societal, health, safety, privacy, security, legal and cultural issues, and the consequent responsibilities relevant to software engineering application	Fundamentals of Database Systems =4/31*49% =6items	4	• Explain fundamentals of database system concepts, technology, models of database and optimization.	1		1							6
				• Write SQL statements for data creation and manipulation purposes	2			2						
				• Design the database models from specifications using a popular DBMS	2						2			
				• Manage the possible problems that may occur in securing data.	1						1			
	= Understand the impact of software applications in societal and environmental contexts and apply the principle of green computing for sustainable development. = Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.	Operating Systems =4/31*49% =6items	4	• Recall the structure and functionality of modern Operating Systems	1	1								6
				• Explain conditions that lead to deadlock and memory management, I/O, mass storage and file management issues	2		2							
				• Describe security and protection measures used in operating systems	1		1							
				• Use knowledge of process management and synchronization for software development	1			1						
				• Compare the various algorithms on which the core functions of the Operating Systems are built on	1				1					

Software requirement, Design, and Architecture =10/64% =16%	=Identify, formulate, research literature, and analyze complex problems, reaching substantiated (justifiable, validated) conclusions using software development principles, methodologies, and tools.	Fundamentals of Software Engineering =4/10*16% =6items	4	• Identifies the basic principles of Software Engineering	1		1							6
				• Describe software process and state-of-the-arts software process models	1		1							
				• Explain fundamental concept of data communication, computer networks, and computer system security as well as setup and configure small-area network.	1		1							
				• Demonstrate installation and configuration of system software, and make use of software engineering tools.	2			2						
				• Be aware of the professional, ethical, and norm of software engineering practice.	1							1		
	Communicate effectively on complex business and system activities with the software engineering community and with a society at large; such as being able to comprehend and write effective reports, articles and design documentations using various tools, make effective presentations and give and receive clear instructions.	Software Requirement Engineering =3/10*16% =5items	3	• Describe the role of requirement engineering in software development	1		1							5
				• Use the appropriate requirements elicitation techniques to identify requirements.	1			1						
				• Develop effective functional and non-functional requirements document that are complete, concise, correct, consistent, testable and unambiguous.	2						2			
				• Manage the change to software requirements.	1						1			
	=Design integrated and efficient software solutions, component, or processes to	Software Architecture and Design =3/10*16% =5items	3	• Explain the role of software architecture and software design principles	1		1							5
				• Apply the different types of design patterns in software development	1			1						

	address complex problems and implement them to meet the specified needs with appropriate consideration for public health and safety, cultural, societal (local and international) and environmental considerations.			• Evaluate various architectural styles and solutions based on quality attributes	1					1				
				• Design a large-scale software system that reflects and balances the different needs of its stakeholders	2						2			
Software project management, maintenance and quality assurance = 9/64*100 = 14	= Demonstrate knowledge and skill on software project management principles and apply these to manage software development processes and its resource requirements, costs in multidisciplinary projects = Function effectively as an independent software engineer, as a member or as a leader in a diverse team and multidisciplinary settings.	Software Project Management =5 items	3	• Explain software project management tasks.	1		1							5
				• Develop work breakdown structure, schedule and cost for software projects.	2						2			
				• Use project management tools, techniques, and skills	1			1						
				• Prepare plan documents for software project management, risk management, software quality management and assurance	1						1			
	=Recognize the need for and have the preparation as well as ability to engage in an independent and lifelong learning in the information and knowledge	Software Testing and Quality Assurance =5 items	3	• Apply modern software testing processes in relation to software development and project management	1			1						5
				• Create test strategies and plans, design test cases, prioritize and execute them.	2						2			
				• Manage incidents and risks within a project	1						1			

	economy.			• Become competent expert in designing, implementation and development of Quality software systems and IT processes	1							1		
	= Apply knowledge of mathematics, and software engineering specialization to address complex and large-scale problems.	Software Evolution and Maintenance =5 items	3	• Examine the major software defects, bugs and code smells	1					1				5
				• Estimate the maintenance cost, efforts, size and time for software maintenance.	1				1					
				• Apply the best practices and models in software maintenance process	1			1						
				• Maintain a software based on the reengineering and refactoring methods and principles.	2						2			
Networking and Software Security =7/64*100 =10.9%	Apply logical reasoning and informed decision to assess societal, health, safety, privacy, security, legal and cultural issues, and the consequent responsibilities relevant to software engineering application	Fundamentals of Networking =4/7*10.9% =6items	4	• Describe concepts of network fundamentals.	1		1							6
				• Analyze the different networking layers	3				3					
				• Design appropriate computer network services based on a given requirement	2						2			
	Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.	Software and information security =3/7*10.9% =5items	3	• Identify malicious code functions, threats, vulnerabilities possible propagation, and mitigation methods	2		2							5
				• Assess computing cryptography and cryptanalysis	2					2				
				• Develop mechanisms for authentication and access control in a domain.	1						1			
				• Compare types of firewall and intrusion detection methods	1				1					
AI and related Areas	Adopt, innovate, select, and apply	Fundamentals of Artificial	3	• Explain fundamental concepts of artificial intelligent agents	1		1							

=6/64*100 =9.4%	appropriate state of the art software design and development tools, methodologies, resources, and emerging technologies including simulation, prediction and modeling for complex business process with understanding of their potential capabilities, limitations and associated risks.	Intelligence =3/6*9.4% =5items		• Compare the basic search strategies used in an intelligent agent.	2				2					5
				• Apply the law of logic to represent knowledge for an expert system	1			1						
				• Engage in independent and lifelong learning with the broad application area of Artificial Intelligence.	1							1		
	=>Conduct investigation of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions	Intr. To Machine Learning =3/6*9.4% =5items	3	• Explain the distinction between supervised and unsupervised learning, as well the interests and difficulties of both approaches.	2		2							5
				• Implement the major machine learning algorithms for data analytics	2						2			
				• Formulate research on a personal project using machine learning techniques	1						1			

Key: In the cells/boxes there are numbers that indicate the number of items that correspond to the levels of domains.

6. Conclusion

Exit examination can have a vital role in producing knowledgeable, skillful and attitudinally matured graduates. It contributes to prepare competent graduates as it can serve as a quality check for effectiveness. It also helps in improving academic programs quality and effectiveness. Furthermore, it can create the platform for cooperation among academic programs at different universities to work jointly to improve the programs quality.

In view of this, this document is produced to assist the setting of the exit examinations for all programs, which is being delivered by Ministry of Education. To meet the graduation profile, competency and learning outcome, exit exam competency selection and identifying core course is done for 2016 EC graduating students. As a result, preparing test blueprint is necessary to prepare fairly distributed items based on the above criteria.

Appendix A:

Table 3: Calculation of share of themes, courses and items from a course from a total of 100 test items for one program

Themes	Course Name	Learning outcomes	Credit hour	Weight of course or proportion	Number of test items from each course	**Cognitive**						**Affective**	**Psychomotor**	Total
						Remember	Understand	Apply	Analyze	Evaluate	Create			
Theme One Share = 49 items from the total	1		8	8/31 = 0.26	0.26*49 = 11	-	1	3	2	-	4	1	-	11
	2		4	4/31 = 0.13	0.13*49 = 6	1	1	2	1	-	1	-	-	6
	3		3	3/31 = 0.1	0.1*49 = 5	-	1	2	-	-	2	-	-	5
	4		6	6/31 = 0.194	0.194*49 = 9	-	2	2	-	1	4	-	-	9
	5		3	3/31 = 0.1	0.1*49 = 5	1	-	2	-	1	-	1	-	5
	6		4	4/31 = 0.13	0.13*49 = 6	-	1	2	-	-	3	-	-	6
	7		4	4/31 = 0.13	0.13*49 = 6	1	3	1	1	-	-	-	-	6
	Theme one total credit		**31**											
Theme Two Share = 16 items from the total	1		4	4/10 = 0.4	0.4*16 = 6	-	3	2	-	-	-	1	-	6
	2		3	3/10 = 0.3	0.3*16= 5	-	1	1	-	-	3	-	-	5
	3		3	3/10 = 0.3	0.3*16 = 5	-	1	1	-	1	2	-	-	5
	Theme two total credit		**10**											
Theme Three Share = 15 items from the total	1		3	3/9 = 0.333	0.333*15 = 5	-	1	1	-	-	3	-	-	5
	2		3	3/9 = 0.333	0.333*15 = 5	-	-	1	-	-	3	1	-	5
	3		3	3/9 = 0.333	0.333*15 = 5	-	-	1	1	1	2	-	-	5
	Theme three total credit		**9**											
Theme Four Share = 11 items from the total	1		4	4/7 = 0.57	0.57*11 = 6	-	1	-	3	-	2	-	-	6
	2		3	3/7 = 0.43	0.43*11 = 5	-	2	-	-	2	1	-	-	5
	Theme three total credit		**7**											
Theme Five Share = 10 items from the total	1		3	3/6 = 0.5	0.5*10 = 5	-	1	1	2	-	-	1	-	5
	2		3	3/6 = 0.5	0.5*10 = 5	-	2	-	-	-	3	-	-	5
	Theme three total credit		**6**											
Total for the program			64			3	21	22	10	6	33	5	-	100

Note: The number of items assigned in the levels of domains is based on the nature of learning outcomes/objectives.

Appendix B: See Curricula of B.Sc. in Software Engineering Program of AASTU and AMU.